

Customer Churn & Engagement Analytics System

Problem Statement

A subscription-based digital platform is facing customer churn and inconsistent engagement, but lacks a unified analytical system to answer:

Why are customers churning?

Which users are highly engaged vs inactive?

Do upgrades and support interactions influence retention?

How do subscription types impact revenue and churn?

Data exists across multiple tables (Customers, Subscription, Usage, Support, Churn)

Project Goal

To build a data analytics solution that integrates customer data across multiple sources and delivers:

A 360° customer view

Insights into engagement, retention, and churn

Identification of high-value and at-risk customers

Data-driven recommendations to improve customer retention and revenue

Key Tasks

Beginner (Data Understanding & EDA)

Goal: Understand customer behaviour and basic trends

Tasks

1. Data Integration

- Load all 5 tables using Pandas
- Merge tables on CustomerID
- Create a unified dataset

2. Data Cleaning

- Handle missing values
- Convert date columns:

- JoinDate, LastLoginDate, ChurnDate
- Remove inconsistent or duplicate records

3. Feature Engineering

- Create:
 - CustomerTenure = Today - JoinDate
 - DaysSinceLastLogin
 - IsChurned (from Status)
 - RevenuePerCustomer

4. Exploratory Analysis

- Total customers vs churned customers
- Churn rate by:
 - Gender
 - City
 - SubscriptionType
- Avg watch hours vs churn
- Support tickets vs churn

5. Visualization

- Bar charts:
 - Churn by subscription
- Line charts:
 - Customer growth over time
- Pie chart:
 - Payment methods distribution

Outcome

- Basic understanding of:
 - Who churns
 - Engagement patterns

- Clean dataset ready for advanced analysis

Intermediate (Customer Segmentation & Insights)

Goal: Identify customer segments and behaviour patterns

Tasks

1. Advanced Feature Engineering

- Create:
 - Engagement Score (WatchHours + Ratings)
 - Support Intensity (tickets per month)
 - Subscription Value Tier

2. Customer Segmentation

- Segment customers into:
 - High Engagement
 - Medium Engagement
 - Low Engagement

3. Behavioural Analysis

- Compare:
 - Engagement vs churn
 - Subscription type vs retention
 - Upgrade offered vs accepted

4. Cohort Analysis

- Group customers by JoinDate
- Track retention over time

5. Visualization

- Cohort retention heatmap
- Revenue by segment

6. Business Insights

- Identify:

- High-value customers
 - At-risk users
- Recommend:
 - Targeted campaigns
 - Subscription improvements

Deliverables

- Unified dataset
- EDA notebook
- Dashboard (Power BI)
- GitHub project